Advance Python Assignment

Topic-6: Class and Object (OOP Concepts)

Q1. Explain class, object, attributes, and methods in Python.

Ans.
In Python, a class is a template used to create objects. It defines the structure and behavior of an object. An object is an instance of a class and represents real-world entities like a student or a car. Attributes are variables that store data inside a class, and methods are functions that define actions performed by the object. These concepts help in organizing the code and make it reusable. Object-oriented programming makes programs more structured and easier to manage.

Example:
```
class Car:
    def __init__(self, brand):
        self.brand = brand

    def display(self):
        print("Brand:", self.brand)

c1 = Car("Toyota")
c1.display()
```

Q2. Explain local and global variables in Python.

Ans.
Local variables are defined inside a function and can be used only within that function. They are created when the function starts and destroyed after the function ends. Global variables are defined outside any function and can be accessed from anywhere in the program. Local variables help in temporary data storage, while global variables are useful for values that need to be used throughout the program. Understanding both helps in writing better and error-free programs.

Example:
```
num = 100  # Global variable
```

```
def test():
    value = 20  # Local variable
    print("Local value:", value)
    print("Global value:", num)

test()
print("Global outside function:", num)
```